



Composite Restoration for a CLASS II cavity preparation

Learning Outcomes



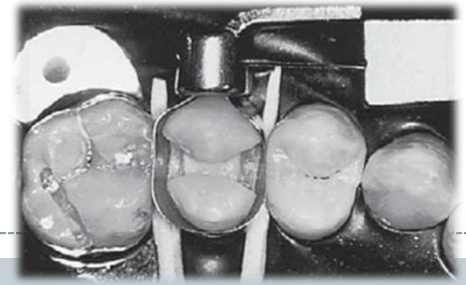
1. Describe in detail about restorative procedure for composite in a class II cavity preparation
1. Describe about curing lights used in composite

Steps in class II preparation and Composite restoration

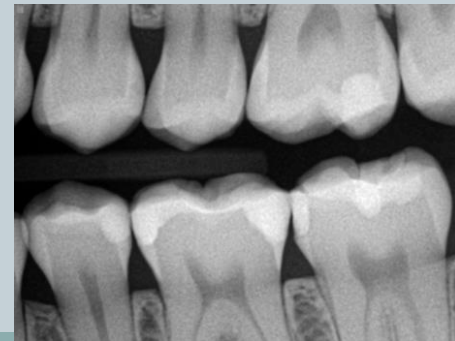
Two step Etch and Rinse technique

- 1) Shade selection
- 2) Isolation of the operating site by rubber dam
- 3) Class II Cavity preparation
- 4) Matrix and wedge application
- 4) Acid etching and wash
- 5) Adhesive application and light curing
- 6) Inserting the composite by incremental layering and light curing
- 7) Finishing & polishing of the composite.

Matrix Application



- One of the most important steps in restoring posterior teeth with composite is the selection and proper placement of the matrix and wedge.
- A properly contoured matrix and wedge is a prerequisite for a restoration involving the entire proximal contact area.





- In contrast to amalgam, which can be condensed to improve the proximal contact development, posterior composites are almost totally dependent on the contour and position of the matrix for establishing appropriate proximal contacts.



Tofflemire matrix



- Before placing the composite material, the matrix band must be in absolute contact with the adjacent contact area.
- Select a matrix band of sufficient height to reach below the cervical margin of the preparation while also extending slightly beyond the marginal ridge.





- Clear polyester matrices are not recommended for larger posterior composite restorations, because they cannot be contoured and they do not provide adequate resistance to condensation if a viscous composite is used.
- Although a Tofflemire-type matrix band can be used for restoring a two-surface tooth preparation, precontoured sectional metallic matrix are preferable



Sectional matrix systems also are available for Class II composite restorations.

- These systems may use a bitine ring to
 - (1) Aid in stabilizing the matrix band and
 - (2) Provide additional tooth separation while the composite is inserted.

The primary benefit promoted by the manufacturers is a simpler method for establishing an appropriate composite proximal contact.

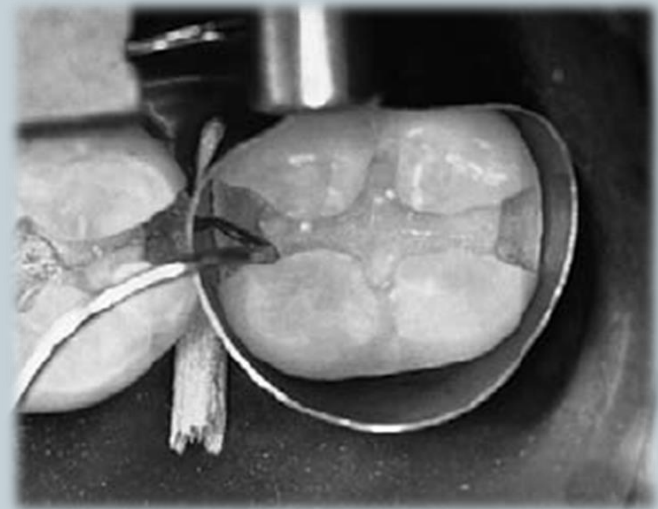




- Regardless of the type of matrix system used, the matrix material should extend at least 1 mm beyond the gingival margin (gingivally) and the area corresponding to the marginal ridge of the restoration (occlusally).
- For posterior restorations, the wedge is placed from the larger to the smaller interproximal gingival embrasure, typically from a lingual approach, just apical to the gingival margin. The wedge should stabilize the matrix band and prevent gingival overhang



Use explorer tip (with pressure) between matrix band and gingival seat to ensure proper adaptation of the band to gingival margin.



Etching procedure

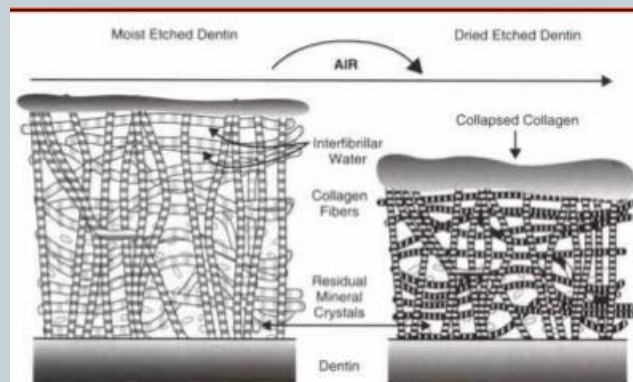


- First, the proximal surface of the adjacent unprepared tooth should be protected from inadvertent etching by placing a tofflemire matrix band or sectional matrix and wedge and if needed protect adjacent teeth with Teflon tape.
- The etchant typically is left undisturbed for 15 seconds. The area is washed to remove the etchant.





- If dentin is exposed, rather than air dry the rinsed area, it may be better to use a damp cotton pellet or a paper tissue to remove the excess water.
- Do not overdry nor pooling of excess water should be allowed

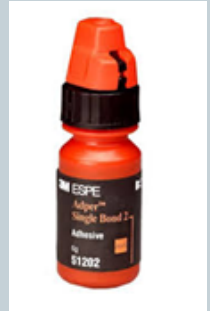




Adhesive application



- Microbrush or applicator tip is used to place the adhesive on all of the tooth structure that has been etched.
- When applied, the adhesive is lightly dried with the air syringe to evaporate any solvent (acetone, alcohol, or water).
- Then the adhesive is polymerized with light curing for 20 seconds .



Inserting and Curing the Composite.



- Nowadays Most composite materials used for restorative purposes are the light cured type.
- The operator should not dispense the bonding adhesive and the composite until they are ready to be used.
- Both of these materials begin to harden when exposed to daylight or other lights in the operator.



- Light cure composites were introduced to overcome the disadvantages of self curing composites

Disadvantages of self curing composites

1. Intrusion of voids.
2. Less working time.
3. Difficulty of manipulation.



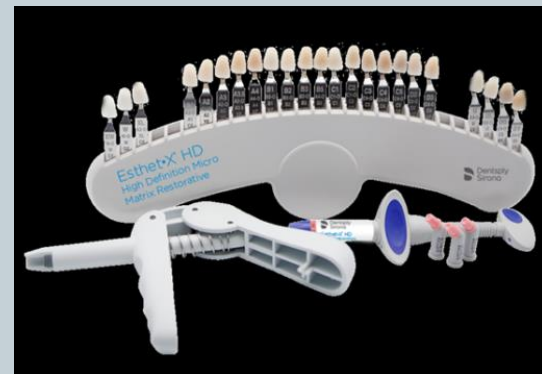
Self cure composite

LIGHT CURED COMPOSITES

- Light-cured composites are available in two forms:
 1. A **threaded syringe** for hand instrument insertion
 2. A self-contained **compule** that is placed into an injection syringe for syringe insertion.



Threaded syringe



Compule System

Threaded syringe for hand insertion

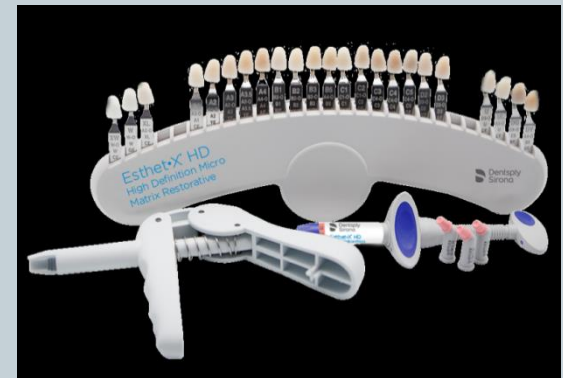


- If a hand instrument is to be used for insertion, the knurled knob on the back of the syringe is rotated to force material out the other end and it is placed onto a paper pad.



Injectable Composite compules

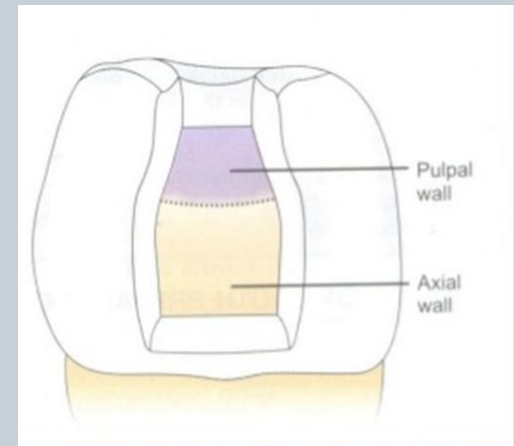
- If the composite is to be injected directly into the preparation, the selected compule is placed into the injection syringe, the protective cap on the compule is removed, and the composite is injected into the preparation





- The protective cap should be replaced when the compule is not being used to protect the material from exposure to ambient light.

- **Whichever the technique is used**, the first small increment should be placed along the gingival floor of proximal box and later should extend slightly up the facial and lingual walls .

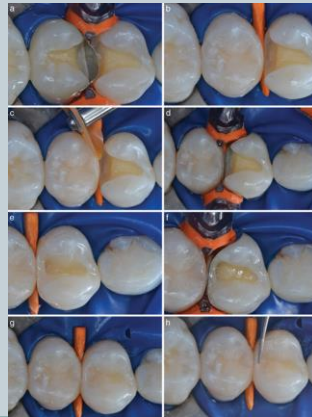




- This increment should be only approximately less than 2 mm in thickness because it is the farthest increment from the curing light and is most critical in establishing a proper gingival seal.
- This first increment should be cured with a light exposure of 20 seconds



- A second increment is then placed against the lingual (or facial) wall to restore about two thirds of the box. The final increment is then placed to complete the proximal box and develop the marginal ridge.
- Ensure the marginal ridge contact point should be the same height as the adjacent ridge of adjacent teeth



- Subsequent additions are made and cured (usually not exceeding 2mm in thickness at a time) until the preparation is completely filled with composite
- With large restorations, it is better to add and cure the composite in several increments to reduce the effects of polymerization shrinkage and to ensure more complete curing in remote regions.



- When a packable high-viscosity composite is used for the restoration of the proximal box, a very small increment of a flowable composite may be first placed in close proximity to the external margins of the proximal box so as to improve marginal adaptation of the restoration.

Removal of matrix and wedge

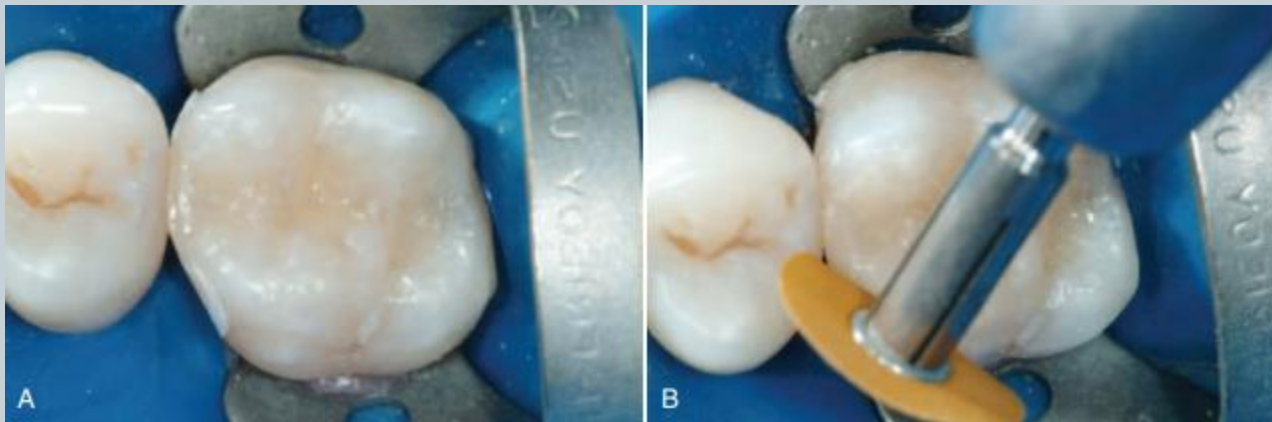


- First remove the matrix band and then wedge
- After matrix is removed, and the restoration is cured from facial and lingual directions.

Contouring and Polishing the Composite

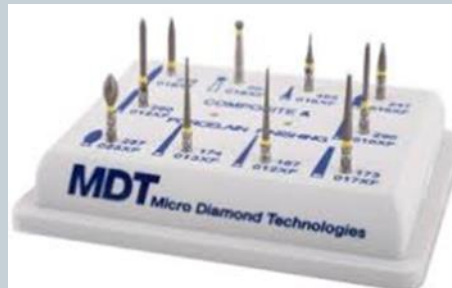


- Finishing can be initiated immediately after a light-cured composite material has been polymerized.
- Excess composite is removed at the proximal margins and embrasures with a flame shaped, 12-bladed carbide finishing bur or finishing diamond and abrasive discs





- The occlusal surface is shaped with a round or oval, 12-bladed carbide finishing bur or finishing diamond.



- Any overhangs at the gingival(proximal)area is removed by Narrow finishing strips may be used to smooth the gingival proximal surface



- The rubber dam is removed, and the occlusion is evaluated for proper contact.



- The proximal surface contours and margins should be assessed visually and tactilely with an explorer and dental floss.
- By passing the floss in between the contact, one should hear snap sound and “shoe-shined” as it is pulled occlusally.
- If the floss catches or frays, additional finishing is indicated.



- Further adjustments are made if needed, and the restorations are finished with appropriate polishing points, cups, spirals .



Light curing units



- It is an instrument capable of generating and transmitting a high intensity blue light with a wavelength oscillating between 400-500 nm that is designed specifically to polymerize visible light sensitive dental material such as composite.





- Photoinitiator in composite – Camphoroquinone (CQ) absorbs blue light with wavelengths between 400-500 nm.

Curing of composites



Four Types of light curing units are

1. Quartz tungsten halogen(QTH)
2. Light emitting diode(LED)
3. Plasma arc (PAC)
4. Argon laser

Quartz tungsten halogen(QTH)



- QTH devices were the widely used light curing units
- It has quartz bulb with tungsten filament in a halogen environment
- QTH curing light works at a wavelength of 400 to 500 nm
- A small fan is inside to dissipate heat from the reflectors
- Orange filters absorb blue radiation and protection of eyes



Advantages :

1. Easy to use
2. Not expensive



Disadvantages :

1. The units are large
2. Light intensity output decreases with time, need bulb replacement after some time
3. Generate high temperature
4. Require ventilating fan

Light emitting diode(LED)



- LED devices uses blue light emitting diodes with silicone carbide
- Small, cordless, light weight and battery operated
- Emit light at a specific length of 410 nm



• Advantages

1. No heat generation
2. No cooling fan required
3. Depth of curing higher than QTH
4. No drop in intensity of light
5. Cordless ,light weight



Disadvantages

1. Batteries must be recharged
2. Expensive than QTH

Plasma arc



- Plasma arc are high intensity light curing units
- Light source – fluorescent bulb containing plasma(Xenon gas)
- Shorter exposure time
- An exposure time of 10 seconds is equivalent to 40 seconds from QTH
- Wavelength 370 to 450 nm



• Advantages

1. Short curing time (Less than 5 seconds)
2. High energy output(Power density of 600-2050 mW/cm²)

Disadvantages

1. Expensive
2. Bulb replacement required after sometime
3. Devices are heavy and bulky
4. Filters and fan required
5. More polymerization shrinkage compared to QTH



Argon laser



- Argon laser produces blue light having high intensity 800 mW/cm^2

Advantages

1. Curing time is short 5 to 10 seconds
2. Uniform depth of cure
3. Uniform polymerization

Disadvantages

1. Expensive
2. Units are large and heavy



References



- Art and science of operative dentistry Sturtevant's, 5th edition Page no – 589 – 593, 205 -209